MINJAE KWON

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Research Interests

Engineering nanostructures that shape light and fluidic behavior, guided by computational modeling.

Keywords: Nanofabrication, Plasmonics & Metasurfaces, Electrokinetics, Simulation & Deep-Learning

Education

Korea University (Seoul, Republic of Korea) Sep. 2025 – Present
Master's Student in the Department of Bioengineering
- Advised by Prof. Yong-Sang Ryu

Korea University (Seoul, Republic of Korea) Mar. 2023 – Aug. 2025
B.E. in Biomedical Engineering & B.S. in Electrical Engineering (Double Major)
- GPA: N.A / 4.50

Research Experience

Ryu Group, Korea University (Seoul, Republic of Korea) Aug. 2024 – Present
Undergraduate/Graduate Student Researcher (Advised by Prof. Yong-Sang Ryu)
- **Current work: MIM Nanohole Plasmonic Metasurface for Single-Particle Analysis (in preparation)**
  Fabrication: Fabricating a single-nanoparticle-capture nanohole metasurface by laser interference lithography, with a metal–insulator–metal stack that enables electrokinetic trapping at SERS hotspots.
  Simulation: Modeling optical and electrokinetic fields to find geometries giving up to ~2×10^6 $|E/E_0|^4$ enhancement and to analyze DEP/ACEO single-particle capture dynamics.
  CNN Analysis: Applying CNNs to SEM images to quantify the relative standard deviation (RSD) of single-nanoparticle capture to analyze biosensor uniformity and the resulting SERS signal.
- **Collaborative projects**
  PRET metasurface: Depositing electrokinetic capture structures onto the Plasmon Resonance Energy Transfer (PRET) metasurface built by Prof. Junsuk Rho's group to concentrate target analytes in aqueous solution.
  Hydrogen-sensing MEMS: Providing deposition and lithography support for a terahertz-resonant MEMS chip for label-free hydrogen sensing.
  Spatial manipulation of phase separation on lipid membrane: Depositing substrates with controlled surface properties to localize liquid-ordered (Lo) domains and spatially control phase separation.
  KU–KIST joint project: Serving as a collaborating researcher at the Nanophotonics Research Center of KIST, developing a label-free plasmonic metasurface for sensing GPCR conformational dynamics.

Creative Challenger Program, Korea University Aug. 2023 – Feb. 2024
Project Leader, Vine Robot Colonoscopy Project (Advised by Prof. Daehee Hong and Dr. Sang-Hyun Kim)
- Developed a soft-robot colonoscopy platform and validated it in vivo with Korea University Medical Center.
- Recognized with two Excellence Awards (15th Creative Challenger Program, 2nd Changemakers Forum)

Publications

[3] **M. Kwon**‡, S. Lee‡, S. Choi, D. Pyun, D.-H. Kim, M.-K. Kim*, and Y.-S. Ryu*, *"**Scalable nanohole plasmonic metasurface for electrokinetic single-particle capture and SERS analysis**,"* manuscript in preparation (2026).

[2] S. Choi‡, S. Lee‡, H. Kim‡, J. Kwon, D. Pyun, T. Kim, **M. Kwon**, S. Jeong, H. S. Song, E.-S. Yu*, M.-K. Kim*, J. Rho*, and Y.-S. Ryu*, *"**Electrical tweezing over Self-SERSitive metasurface enables 1 pg/L Nanoplastic detection**,"* under review, Nature Communications (2026).

[1] S. Lee‡, S. Choi‡, T. Kim, **M. Kwon**, T. J. Mun, S. Son, H. Seong, K. Lee, H. Im, E.-S. Yu, M.-K. Kim*, and Y.-S. Ryu*, *"**Analyte-accessible field volume as a governing design principle for nanophotonic sensing**,"* under revision, Advanced Materials (2026).

‡Equal contribution, *Corresponding author

Technical Proficiency

Fabrication & Characterization Skills

- Lithography: Photolithography, Laser Interference Lithography (LIL), Electron Beam Lithography (EBL), Soft Lithography, Nanoimprint Lithography (NIL)
- PVD (Physical Vapor Deposition): Thermal Evaporation, Electron Beam Evaporation, Sputtering
- CVD (Chemical Vapor Deposition): ALD, PECVD
- Etching: RIE, ICP-RIE
- Surface Treatment: Oxygen Plasma Ashing, UV-Ozone
- Material / Structure Analysis: SEM, AFM, Energy Dispersive X-ray Spectroscopy (EDS), Surface Profiler, Ellipsometry, Reflectance Spectroscopy, Raman Spectroscopy, Optical & Epi-fluorescence Microscopy

Computational & Design Skills

- Language: Python, C/C++, LaTeX, MATLAB
- Simulation: COMSOL Multiphysics (FEA), Lumerical FDTD
- Data Analysis: OriginLab, Veusz, Python-based data analysis (PyTorch, OpenCV)
- 2D/3D Design: Rhino, AutoCAD, KeyShot
- Physical Design: PCB/Breadboard Circuit Design, 3D Printing, Metal / Polymer CNC, Laser Cutting

Teaching Assistant

Biomedical Semiconductor Process (BMED442, Prof. Yong-Sang Ryu) — Fall 2025, Fall 2026
Biomedical Design (HBL514, Prof. Yong-Sang Ryu) — Spring 2026

Awards and Scholarships (Selected)

BK21 FOUR Graduate Research Scholarship, Ministry of Education (Full Tuition) — Sep. 2025 – Present
Best Poster Presentation Award, Korean Biochip Fall Symposium — Nov. 2025
Outstanding New Student Scholarship, Korea University — Sep. 2025
Dean's List (Highest Honors, 2 semesters), Korea University — Mar. 2023 – Jun. 2025
Excellence Award, 2nd Changemakers Forum — Nov. 2024
Best Startup Award, Korea University — Nov. 2024
Third Place, Capstone Design Fair — Sep. 2024
Excellence Award, 15th Creative Challenger Program — Feb. 2024

Leadership & Extracurricular Activities (Selected)

School of Biomedical Engineering, Korea University

Founder & President, KU Biomedical Eng. Abroad Mentorship (KUBEAM) — Aug. 2025 – Present

- Founded a mentorship initiative connecting 80+ students and alumni pursuing study/careers abroad, building a shared study-abroad resource hub and overseas alumni network.

Biosensor Team Lead Organizer, Biomedical Engineering Academic Team (BEAT) — Aug. 2025 – Jul. 2026

- Led and mentored 60+ undergraduates in acquiring the concepts and technical skills necessary to explore the field of biosensors (e.g., SEM imaging, data analysis, critical literature review, presentation).

Senior Advisor & Vice President, Department Student Council — Nov. 2024 – Nov. 2025

- Advised and supported planning and execution of academic, cultural, and community events for 500+ undergraduates. Led a 30+ member committee and secured $22,000 in sponsorships for a 500+ student alumni day.

65th Medical Brigade, 8th U.S. Army — Jul. 2019 – Jan. 2021

Combat Medic & Medical Support Assistant, BDAACH (549th Hospital Center)

- Selected for the KATUSA (Korean Augmentation to the U.S. Army) program based on language proficiency. Served as staff in the Physical Performance Service Line (musculoskeletal rehabilitation and physical therapy) for outpatient care and on the COVID-19 Response Team (quarantine operations, vaccine administration).
- Served as Senior KATUSA of 65th Medical Brigade, the ranking KATUSA representative and liaison to U.S. command; received the Army Achievement Medal; honorably discharged as Sergeant (E-5).